LIAM PACKER

lp492@cornell.edu ◇ liampack.github.io

EDUCATION

Cornell University *Ithaca, NY* **August 2023–Present**
PhD Candidate—Applied Mathematics (Expected Graduation: May '28)

Swarthmore College *Swarthmore, PA* **Aug 2016–May 2020**
Bachelor of Arts—High Honors with majors in Mathematics and Physics

WORK EXPERIENCE

Johns Hopkins University Applied Physics Laboratory *Laurel, MD* **Jun 2020–Jun 2023**
Scientific Analysis, Machine Learning, and Software Design

SES/SRN: Space Analysis and Applications Group (Feb 2022–Jun 2023):
The SRN group of the space sector is a research group focused in aiding research across all levels of technology, from theoretical underpinnings to flight-ready design. Primarily NASA-sponsored.

AOS/QAT: Tactical Intelligence Systems Group (Jun 2020–Feb 2022):
The QAT group of the Asymmetric Operations sector focuses in building intelligent systems for flight-ready situations. Primarily sponsored by the DoD's Chief Digital and Artificial Intelligence Office (CDAO).

AWARDS AND FELLOWSHIPS

Cornell Knight Institute Buttrick-Crippen Assistantship Award **Fall 2026–Spring 2027**
Cornell Math Teaching Development Fellow **Spring 2026**
Cornell University Graduate Resident Fellow **Jun 2024–Present**
NSF Graduate Research Fellowship Program (HM) **Spring 2025**
Swarthmore College Summer Research Fellowship **Summer 2019**
Swarthmore College Summer Research Fellowship **Summer 2018**

PUBLICATIONS AND PREPRINTS

[1] Liam Packer, Kihoon Seong, and Phil Sosoe. Large Deviation Principles for the Gross-Pitaevskii Gibbs Measure at Low Temperature. *arXiv preprint arXiv:2510.13206*, 2025.

TEACHING

TA Training (Math 6080) *Cornell University* **Spring 2026**
Ran micro-teaching workshops to provide first-year graduate students feedback on recitations.

Teaching Assistant (Math 6110) *Cornell University* **Fall 2025**
Graduate core course in analysis. Teaching advanced topics weekly in recitation, office hours, and grading.

TA Training (Math Dept) *Cornell University* **Aug 2025**
Instructing on roles and responsibilities for Cornell math TAs.

Instructor (UNLIWYL 1407) *Cornell University* **Spring 2025**
Course on academic communication, co-taught with house professor and peer resident fellow.

Teaching Assistant (Math 4710) *Cornell University* **Fall 2024**
Upper-level introductory probability. Weekly recitations, office hours, and grading.

Teaching Assistant (Math 2240) *Cornell University* **Jan 2024–May 2024**
Honors-track course modeled after Harvard 55a/b. Weekly recitations, office hours, and grading.

Teaching Assistant (Math 1920) *Cornell University* **Fall '23, Spring '25**
Held weekly recitations for ≈ 75 students. Prepared quizzes and section materials from scratch.
Math Clinician *Math Department, Swarthmore College* **Sep 2019–May 2020**
Walk-in mathematics peer tutor for over a hundred students, in all math classes offered by the department.
Teaching Assistant *Physics Department, Swarthmore College* **Dec 2018–May 2020**
Held weekly three-hour office session for students in Mechanics, E&M, Thermodynamics, and Optics.
Algorithms Grader *Computer Science Department, Swarthmore College* **Fall 2018, Fall 2019**
Assessed ≈ 80 pages of proof-based work weekly for the advanced algorithms class.

MENTORING

Directed Reading Program *Cornell University* **Aug 2024–Present**
Julia Robinson Math Festival *Cornell University* **Spring 2025**
CAM Mentoring Program Coordinator *Cornell University* **Aug 2024–Aug 2025**
Graduate Resident Fellow *Cornell University* **Aug 2024–Present**
Cornell Math Modeling Judge *Cornell University* **Fall '23, '24, '25**
Association for Women in Mathematics Mentor *Cornell University* **Nov 2023–Present**
Intern Advisor *Johns Hopkins Applied Physics Lab* **Jun 2022–Jun 2023**

PRESENTATIONS AND TALKS

Taxonomies of Knowledge in Math Education — *Cornell, Math Teaching Seminar* **Feb 2026**
Uniqueness of the ϕ_2^4 field via Holley-Stroock — *Cornell, PhD Candidacy Exam* **Dec 2025**
Langevin Dynamics and Heat Semigroups — *Cornell, CAM student seminar* **Oct 2024**
Percolation on $\mathbb{Z}^2$ and Kesten's theorem — *Cornell, CAM student seminar* **May 2024**
Kalman Filters and Hilbert Spaces — *Cornell, Applied Dynamics Seminar* **Feb 2024**
Kalman Filters and Hilbert Spaces — *Cornell, Sarah Dean Group* **Feb 2024**
Spectral Graph Theory for Clustering in CRISM MSV data — *SES/SRN* **Oct 2022**
CV Techniques for Unsupervised Clustering of Hyperspectral Imagery — *SES/SRN* **Aug 2022**
SAR Sensor Planning: Optimizing Pointing Decisions for Unusual Sensors — *AOS/QAT* **Sep 2021**
Neuro-symbolic Methods in Image-to-Text Generation — *AOS/QAT* **Sep 2020**
Jammed solids held together with pins: structure and dynamics — *Swarthmore College* **Dec 2019**
Predicting Cluster Memory Usage for Adaptive Network RAM — *Swarthmore College* **Aug 2018**

SUMMER SCHOOLS ATTENDED

Summer School in PDE's and Randomness at MPI MiS **Summer 2026**
École d'Été de Probabilités de Saint-Flour LIV **Summer 2026**
SLMath: Random Growth Models, Phase Separation and Hamilton-Jacobi PDE **Summer 2026**
Cornell Probability Summer School **Summer 2024**

TECHNICAL PROFICIENCIES

Computer Languages	C++, Python, Julia, C, Scheme, Racket, Java, OCaml
Mathematical Computing	Maxima, LaTeX, IDL, Matlab
Software & Frameworks	Emacs, Linux, Pytorch, Scikit-Learn, ReactJS, Pandas
Tooling & Build Systems	Git, Docker, Tmux, Maven, Make, CMake, Dune
Languages	Japanese & Spanish (limited working proficiency)

EXTRACURRICULAR

- Co-president of *Kizuna*, the resident Japanese Culture group at Swarthmore College (2017–2019).
- President of Swarthmore's Smash club, hosting weekly tournaments with up to 30 entrants (2018–2020).
- Classically trained flutist. Performed in a number of venues including Carnegie Hall.
- Recreating the (computational) wheel in Chez Scheme: monadic parsing, matrix computations, see github.com/liampack.